

MIDLAND INTL, TX, USA

WMO: 722650

Lat: 31.948N

Lon: 102.209W

Elev: 872

StdP: 91.28

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23023

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-4.1	-17.4	0.9	4.2	-15.0	1.1	7.1	14.3	12.4	12.4	11.6	3.8	30	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	38.7	19.5	37.3	19.7	36.1	19.7	22.8	31.4	22.3	30.9	21.9	30.3	5.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	17.2	24.6	20.2	16.6	24.6	19.7	16.0	24.7	71.7	31.5	69.7	30.8	68.1	30.3	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.0	10.8	9.3	DB	-9.9	41.2	2.0	1.9	-11.3	42.6	-12.4	43.7	-13.5	44.7	-14.9	46.1	(4)
			WB	-11.0	23.6	1.8	0.7	-12.2	24.1	-13.3	24.6	-14.3	25.0	-15.5	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	7.3	9.7	13.9	18.6	23.6	27.8	28.5	28.1	24.2	18.8	12.2	7.6	(6)
		DBStd	8.88	4.99	5.63	5.33	4.71	4.32	2.93	2.37	2.43	3.47	4.66	4.94	4.84	(7)
		HDD10.0	348	112	67	24	3	0	0	0	0	3	34	104		(8)
		HDD18.3	1375	342	243	152	53	9	0	0	3	51	188	334		(9)
		CDD10.0	3420	29	60	146	262	422	533	575	560	427	275	101	29	(10)
		CDD18.3	1404	0	2	16	62	173	283	317	302	181	64	5	0	(11)
		CDH23.3	17413	13	69	288	844	2127	3563	4075	3734	1906	690	89	14	(12)
		CDH26.7	9108	1	12	80	337	1109	2019	2311	2068	898	258	15	0	(13)
(14)	Wind	WSAvg	4.9	4.5	4.9	5.3	5.7	5.6	4.8	4.3	4.3	4.6	4.4	4.4		(14)
(15)	Precipitation	PrecAvg	367	14	15	15	19	45	42	43	47	55	39	18	15	(15)
(16)		PrecMax	816	64	65	73	63	194	101	216	165	246	189	139	84	(16)
(17)		PrecMin	130	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	135	15	15	16	19	39	28	50	38	49	44	24	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	25.1	28.5	31.4	34.5	39.0	41.4	39.8	39.3	37.1	33.9	28.8	25.4		(19)
(20)		MCWB	11.2	12.3	13.5	14.4	17.7	19.2	20.2	20.5	19.2	17.4	14.2	11.4		(20)
(21)		DB	22.2	25.5	28.9	32.4	36.8	39.0	37.9	37.6	35.1	31.7	25.7	22.2		(21)
(22)		MCWB	9.9	11.3	13.1	14.5	17.0	19.1	20.3	20.4	19.3	17.4	13.8	10.8		(22)
(23)		DB	19.7	22.9	26.7	30.6	34.9	37.1	36.7	36.3	33.5	29.7	23.7	19.3		(23)
(24)		MCWB	9.0	10.8	12.5	14.2	16.8	19.6	20.3	20.5	19.4	17.0	13.0	9.5		(24)
(25)		DB	16.8	20.1	24.3	28.5	32.8	35.3	35.5	34.9	31.8	27.4	21.4	16.6		(25)
(26)		MCWB	8.2	9.7	12.0	14.1	17.0	19.8	20.3	20.5	19.3	16.4	12.3	9.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	13.9	15.5	17.7	19.5	22.2	23.5	23.3	23.3	22.6	21.8	18.2	15.1		(27)
(28)		MCDB	18.9	21.0	23.2	26.4	29.5	32.5	31.5	32.0	29.7	28.2	21.9	18.5		(28)
(29)		WB	11.8	13.9	16.4	18.5	21.2	22.8	22.6	22.5	21.9	20.6	16.3	12.8		(29)
(30)		MCDB	17.4	19.4	21.8	24.4	28.4	31.5	31.2	31.5	28.8	26.1	21.1	18.1		(30)
(31)		WB	10.2	12.3	15.3	17.5	20.4	22.1	22.1	22.0	21.4	19.7	15.1	10.9		(31)
(32)		MCDB	17.6	19.9	21.3	23.6	27.7	30.7	30.8	31.1	28.3	24.9	20.3	17.6		(32)
(33)		WB	8.7	10.8	13.9	16.4	19.7	21.6	21.7	21.6	20.8	18.6	13.8	9.3		(33)
(34)		MCDB	16.1	18.7	21.1	23.7	27.3	30.2	30.4	30.5	27.8	23.9	19.4	16.2		(34)
(35)	Mean Daily Temperature Range	MDBR	14.5	14.8	15.2	15.5	14.3	13.6	12.6	12.5	12.5	13.5	14.0	14.0		(35)
(36)		5% DB	18.8	19.1	18.1	17.6	16.8	15.9	14.1	14.0	14.7	16.0	17.0	18.4		(36)
(37)		MCWBR	9.3	8.9	7.4	7.0	6.2	4.6	4.0	3.8	4.6	6.1	7.8	9.4		(37)
(38)		5% WB	16.2	15.8	14.7	14.4	14.1	13.1	12.2	12.3	11.4	11.8	13.2	15.4		(38)
(39)		MCWBR	8.8	8.8	7.3	7.2	6.1	4.3	3.6	3.6	3.8	5.4	7.3	8.8		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.265	0.276	0.295	0.331	0.348	0.378	0.396	0.373	0.353	0.314	0.287	0.270		(40)
(41)		taud	2.578	2.548	2.507	2.400	2.415	2.379	2.374	2.413	2.437	2.530	2.540	2.564		(41)
(42)		Ebn at Noon	978	993	990	958	937	904	886	906	915	935	940	952		(42)
(43)		Edh at Noon	79	91	102	118	118	122	122	116	108	91	82	77		(43)
(44)	All-Sky Solar Radiation	RadAvg	3.47	4.30	5.56	6.80	7.21	7.61	7.21	6.69	5.58	4.57	3.64	3.10		(44)
(45)		RadStd	0.34	0.45	0.40	0.42	0.50	0.35	0.40	0.29	0.38	0.50	0.38	0.28		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	+120
Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page